

## MASALALAR YECHISH

55 -  
mavzu

1 Ko'ndalang kesim yuzi  $0,2 \text{ mm}^2$  bo'lgan nikelin o'tkazgich uchlariga  $4,5 \text{ V}$  kuchlanish berilganda undan  $300 \text{ mA}$  tok o'tdi. O'tkazgich uzunligi qanday bo'lgan?

| Berilgan:                                                                                                                                                                 | Formula                                                                                                                                                                                                                                                     | Hisoblash                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $S = 0,2 \text{ mm}^2 = 0,2 \cdot 10^{-6} \text{ m}^2$<br>$U = 4,5 \text{ V}$<br>$I = 300 \text{ mA} = 0,3 \text{ A}$<br>$\rho = 0,4 \cdot 10^{-6} \Omega \cdot \text{m}$ | $R = \rho \frac{l}{S}$ va $R = \frac{U}{I}$<br>$l = \frac{U \cdot S}{\rho \cdot I}$<br>$[l] = \frac{\text{V} \cdot \text{m}^2}{\Omega \cdot \text{m} \cdot \text{A}} = \frac{\text{V} \cdot \text{m}}{\frac{\text{V}}{\text{A}} \cdot \text{A}} = \text{m}$ | $l = \frac{4,5 \cdot 0,2 \cdot 10^{-6}}{0,4 \cdot 10^{-6} \cdot 0,3} \text{ m} = 7,5 \text{ m}$<br><b>Javob:</b> $l = 7,5 \text{ m}$ . |
| <b>Topish kerak:</b><br>$l = ?$                                                                                                                                           |                                                                                                                                                                                                                                                             |                                                                                                                                        |

2 Uzunligi  $20 \text{ m}$  va ko'ndalang kesim yuzi  $2 \text{ mm}^2$  bo'lgan nixrom o'tkazgich uchlariga  $44 \text{ mV}$  kuchlanish berilganda undan qanday tok o'tadi?

| Berilgan:                                                                                                                                                                        | Formula                                                                                                                                                                                                    | Hisoblash                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $l = 20 \text{ m}$<br>$S = 2 \text{ mm}^2 = 2 \cdot 10^{-6} \text{ m}^2$<br>$U = 44 \text{ mV} = 44 \cdot 10^{-3} \text{ V}$<br>$\rho = 1,1 \cdot 10^{-6} \Omega \cdot \text{m}$ | $R = \rho \frac{l}{S}$ va $R = \frac{U}{I}$<br>$I = \frac{U \cdot S}{\rho \cdot l}$<br>$[I] = \frac{\text{V} \cdot \text{m}^2}{\Omega \cdot \text{m} \cdot \text{m}} = \frac{\text{V}}{\Omega} = \text{A}$ | $I = \frac{44 \cdot 10^{-3} \cdot 2 \cdot 10^{-6}}{1,1 \cdot 10^{-6} \cdot 20} \text{ A} = 4 \cdot 10^{-3} \text{ A}$<br><b>Javob:</b> $I = 4 \text{ mA}$ . |
| <b>Topish kerak:</b><br>$I = ?$                                                                                                                                                  |                                                                                                                                                                                                            |                                                                                                                                                             |

3 Cho'g'lanma lampochka spiraling uzunligi  $10 \text{ cm}$  va ko'ndalang kesim yuzi  $0,006 \text{ mm}^2$  bo'lgan volframdan yasalgan. Lampochkaga ketma-ket ulangan ampermetr  $2,4 \text{ A}$  tok kuchini ko'rsatmoqda. Lampochka uchlaridagi kuchlanishni aniqlang.

| Berilgan:                                                                                                                                                                  | Formula                                                                                                                                                                                                              | Hisoblash                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $l = 10 \text{ cm} = 0,1 \text{ m}$<br>$S = 0,006 \text{ mm}^2 = 6 \cdot 10^{-9} \text{ m}^2$<br>$I = 2,4 \text{ A}$<br>$\rho = 0,055 \cdot 10^{-6} \Omega \cdot \text{m}$ | $R = \rho \frac{l}{S}$ va $I = \frac{U}{R}$<br>$U = I \cdot R = I \cdot \frac{\rho \cdot l}{S}$<br>$[U] = \text{A} \cdot \frac{\Omega \cdot \text{m} \cdot \text{m}}{\text{m}^2} = \text{A} \cdot \Omega = \text{V}$ | $U = 2,4 \cdot \frac{0,055 \cdot 10^{-6} \cdot 0,1}{6 \cdot 10^{-9}} \text{ V} = 2,2 \text{ V}$<br><b>Javob:</b> $U = 2,2 \text{ V}$ |
| <b>Topish kerak:</b><br>$U = ?$                                                                                                                                            |                                                                                                                                                                                                                      |                                                                                                                                      |



1 O'tkazgich uchlariga 6 V kuchlanish berilganda undan 5 s da 20 C zaryad o'tdi. O'tkazgich qarshiligi qanchaga teng?

2 Uzunligi 12 m va ko'ndalang kesim yuzi  $0,6 \text{ mm}^2$  bo'lgan nixrom o'tkazgich uchlariga 4,4 V kuchlanish berilganda undan qancha tok o'tadi?

3 Uzunligi 5 m, ko'ndalang kesim yuzi  $0,5 \text{ mm}^2$  bo'lgan simning qarshiligi  $4 \Omega$  ga teng. Sim qanday moddadan tayyorlangan?

4 Qarshiligi  $10 \Omega$  bo'lgan o'tkazgich uchlariga 2,5 V kuchlanish berilgan. O'tkazgich ko'ndalang kesim yuzidan 8 s da qancha elektron o'tadi?

5 Ko'ndalang kesim yuzi  $0,1 \text{ mm}^2$  bo'lgan nixromdan elektr plitasining qizdirgichi yasalgan. Uning uchlariga 220 V kuchlanish berilganda undan 4 A tok o'tdi. Qizdirgichga qanday uzunlikdagi sim olingan?

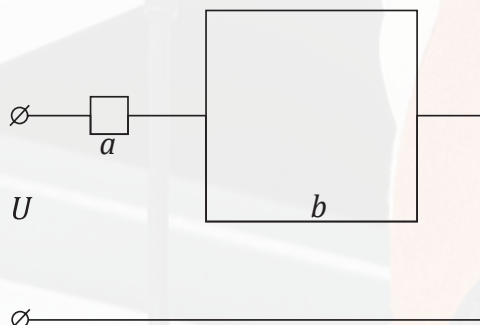
6 Qarshiligi  $1,7 \Omega$  bo'lgan mis simda 0,5 A tok hosil qilish uchun shu simning uchlariga qancha kuchlanish qo'yish kerak? Simning ko'ndalang kesim yuzi  $0,5 \text{ mm}^2$  bo'lsa, uning uzunligi qanday bo'ladi?

7 Uzunligi 100 m, ko'ndalang kesim yuzi  $0,5 \text{ mm}^2$  bo'lgan alyuminiy simning uchlaridagi 7 V kuchlanish berilganda shu simdan o'tayotgan tok kuchi qancha bo'ladi?

8 Maxsus dastgohda simni cho'zib, ikki marta uzun va ingichka qilingan. Natijada simning qarshiligi qanday o'zgargan?

9 Akkumulyatorga ulangan voltmeter 12 V ko'rsatadi. Agar akkumulyator qisqichlaridan biri yerga ulansa, voltmeter ko'rsatishida qanday o'zgarish bo'ladi?

10\*. Qalinligi bir xil tomoni  $a$  va  $b$  ga teng bo'lgan kvadrat shaklidagi ikkita mis plastinka zanjirga 4.66-rasmda ko'rsatilgandek ulangan. Plastinkalarning tokka qarshiligini taqqoslang.



4.66-rasm